

From ECC Semantics to Physical Outcomes: Identity-Preserving Evaluation of Memory ECC Hardware

Anonymous submission

Abstract—An error-correcting-code (ECC) label specifies coding semantics, not a unique RTL architecture or physical outcome. We present an identity-preserving evaluation discipline that qualifies behavior before physical measurement and joins every result to frozen RTL, transaction timing, implementation condition, activity trace, and matched place-and-route seed. For exact-equivalent combinational and pipelined SECDED (72,64) at 10 ns, pipelining changes area by 37.2%, a post-route signed-slack-derived frequency estimate by 46.6%, and steady-stream energy/op by -23.4% over five matched deterministic seeds; request latency changes from one to three cycles. A second exact-equivalent Hsiao pair holds latency and initiation interval fixed while replacing a flat decoder with hierarchical nibble decoding. Its physical displacement is modest and mixed: area -0.564%, detailed wire 3.867%, timing 2.656%, and energy 1.328%. Component-level power evidence attributes the SECDED energy reduction numerically to a -49.48% switching decrease despite higher internal, sequential, clock, and leakage contributions. A prospectively fixed 5 ns replication preserves the temporal pair's area/timing/energy ordering in all five seeds, with condition-dependent magnitudes. These results show that correctness, hardware identity, physical condition, feasibility, and power evidence must remain explicitly joined; implementation identity determines what must be measured separately, not how large every transformation's effect must be.

Index Terms—memory ECC, SECDED, Hsiao, BCH, formal equivalence, physical design, activity power

I. INTRODUCTION

Memory ECCs are commonly named by rate and guaranteed correction/detection behavior. Those properties are essential, but they do not fix encoder/decoder structure, register placement, latency, clocking, routing, or activity. Thus a code label is not a sufficient physical experimental unit: two RTLs may implement the same input/output relation yet occupy different trade-off regions, and neither correctness evidence nor PPA can be transferred merely because labels match.

Hardware-conscious SEC/SECDED construction and checker optimization are established [1]–[4]; multi-error code hardware is likewise architecture-dependent [5]–[9]. Cross-layer studies trade reliability against array, storage, performance, or energy cost [10]–[16]. Our narrower question is experimental: after correctness qualification, how do temporal RTL structure, non-temporal logic structure, physical heuristics, and implementation constraint displace the physical outcomes of frozen ECC hardware identities?

We test the relation

**ECC semantics $\neq$ hardware architecture identity $\neq$
physical implementation identity.**

with predeclared matched OpenROAD seeds. Exact-equivalent combinational and pipelined SECDED isolate temporal restructuring; an exact-equivalent flat/hierarchical Hsiao pair isolates non-temporal decoder organization; a prospectively fixed 5 ns campaign tests constraint sensitivity; component-level power analysis explains the temporal energy difference; and one qualified BCH (78, 64, $t = 2$) realization provides a stronger-correction feasibility boundary.

We organize the evidence around three questions. *Q1*: Does a semantics-preserving temporal transformation retain its physical trade direction across matched heuristic perturbations? *Q2*: When latency and II are also fixed, can a structurally distinct decoder produce a much smaller yet still measurable displacement? *Q3*: Which reported temporal effects survive a tighter implementation condition, and which power components numerically account for the energy ordering? The questions separate effect existence, magnitude, robustness, and mechanism accounting; a positive answer to one does not supply the others.

This paper contributes: (1) a formal-to-physical evidence discipline joining semantics, RTL and transaction identity, implementation condition, feasibility, and activity power; (2) two exact-equivalence architecture studies showing, respectively, large temporal and modest mixed structural displacement; and (3) constraint and power-mechanism characterization, with stronger correction kept categorical. We propose no new ECC and make no family-wide or technology-portability claim.

II. RELATED WORK

A. ECC hardware and cross-layer protection

Hsiao's odd-weight-column construction already makes parity-check structure a hardware concern [1]. Subsequent checker work optimizes Hamming/Hsiao logic for power, area, or delay [2], [3], and ASIC studies compare SECDED with adjacent guarantee choices such as double-adjacent-error correction [4]. These papers establish that code construction and logic organization matter. Our exact-equivalent Hsiao experiment asks a different question: when code, matrix, transaction behavior, latency, and II are fixed, how much does one decoder representation move after a common physical flow? The answer can be small without weakening the need to distinguish the implementations.

Stronger-correction hardware further warns against family-level conclusions. Parallel Chien search trades replicated evaluation logic against latency [6]; low-power variants gate or re-

organize search activity [7], [8]; and recent direct/conventional BCH implementations occupy different area-efficiency and latency regions [9]. Comparative DEC work likewise exposes architecture-dependent cost [5], [13]. We therefore retain the repository’s BCH syndrome/Chien point as one qualified realization. Its target miss is evidence about that identity, not about the feasibility of BCH as a code family.

Memory studies co-design ECC with STT-MRAM, adaptive granularity, GPU register files, DRAM protection, and reliability modeling [10]–[12], [14]–[16]. We instead hold storage arrays and fault rates out of scope and measure codec logic under a common digital flow.

Those cross-layer studies answer system questions that our logic boundary cannot: device error mechanisms, storage overhead, workload adaptation, and operational failure behavior. Conversely, a system-level code label can hide whether compared codec costs came from the same latency, physical condition, or power trace. Our contribution is complementary: it makes the hardware and physical identity of the codec cost explicit so that a later array- or system-level model can attach a measured cost to the correct implementation. We deliberately do not turn exhaustive W1/W2 guarantees into FIT, SDC, or DUE rates without a physical fault distribution.

B. Equivalent representations and physical outcomes

Logic rewriting and learned synthesis-sequence exploration show that equivalent representations and optimization recipes expose different area/delay opportunities [17]–[20]. Physical-flow parameter tuning likewise changes quality of results [21], while OpenROAD and OpenLANE enable inspectable RTL-to-layout experiments [22], [23]. Prior work therefore separately establishes ECC architecture, synthesis sensitivity, and physical optimization. Our focus is their explicit conjunction with formally qualified RTL identity and matched physical perturbations; we do not claim that heuristic-seed study itself is unprecedented. Formal equivalence supplies the behavioral boundary [24], [25], and activity-based power estimation motivates keeping the applied trace and internal switching evidence explicit [26], [27].

The conjunction creates two reporting obligations. First, equivalence must be stated at the correct boundary: combinational arbitrary-word equality and temporally aligned transaction equality are not interchangeable claims. Second, physical variability cannot be addressed by quoting one favorable place-and-route trajectory. Learned or tuned flows may improve quality of results, but changing a synthesis sequence, floorplan policy, or parameter set also changes the physical identity. We instead freeze those choices and use matched seeds only as controlled heuristic perturbations. This sacrifices claims about globally optimal implementation in exchange for a cleaner within-flow contrast.

Activity-based estimation has a parallel identity problem. Average total power without a trace, duration, useful-operation count, and annotation boundary is not an energy comparison. Classical probabilistic and activity-based power work explains why internal transition behavior matters [26], [27]; it does not

authorize a causal glitch claim from aggregate totals alone. Hence our decomposition reports internal, switching, leakage, and cell-group movements, while the mechanism language remains explicitly hypothetical. The literature positioning is therefore intentionally narrow: we contribute neither a new code nor a new synthesis or power algorithm, but an auditable way to join their outputs for implementation-sensitive ECC evaluation.

III. IDENTITY-PRESERVING EVALUATION

A. Three identities and qualification

Figure 1 separates the objects joined by a claim. A *code identity* fixes payload/codeword mapping and decoder semantics. A *hardware identity* adds RTL sources, interface, request latency, initiation interval (II), and structure. A *physical identity* adds constraints, libraries/corner, tools, seed, routed database, parasitics, and activity. PPA cannot qualify behavior; exact behavior does not authorize transferring PPA between hardware identities.

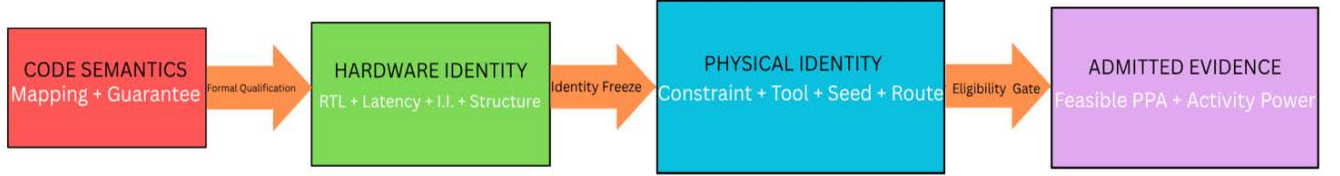
The two extended-Hamming SECDED candidates implement the same systematic (72,64) transaction relation. Formal equivalence after a two-cycle temporal alignment proves identical encoder/decoder responses. Both accept requests at II=1; combinational and pipelined request latencies are one and three cycles.

For Hsiao, the historical table representation exceeded the unchanged synthesis-memory policy, so its unavailable PPA is not inherited. The admitted baseline instead uses explicit syndrome comparisons. The new identity preserves encoder, syndrome matrix, interface, status, one-cycle latency, and II=1, but shares high/low-nibble decode logic. An unconstrained 72-bit miter proves decoder equality for every received word; registered-boundary temporal induction proves same-cycle transaction equality from equal initialization under arbitrary inputs. Zero temporal alignment makes this a structural, not timing, comparison. Common mapping produces 1244 versus 1254 cells, 349 versus 349 registers, and 350 versus 343 XOR/XNOR cells. Histograms and normalized graph signatures differ. Mapped logic depth is *not assessable*: the common report exposed no paths, and absence is not converted into a measurement.

SECDED and Hsiao correct every weight-1 (W1) coordinate error and detect every W2 error in their qualified universes. The shortened BCH corrects W1/W2. Guarantee remains categorical rather than a scalar PPA objective.

B. Physical and measurement contract

The container-pinned ORFS/OpenROAD flow targets SKY130HD at TT, 1.80 V and 25°C. Common settings include 35% utilization, 0.55 placement density, 10 μ m core margin, 0.05 pF output load, 10% period-relative I/O delays, and one worker. The physical seed drives global placement, global routing, and general OpenROAD randomness. The prospectively chosen set is {11, 13, 17, 19, 23}; values are controlled deterministic heuristic perturbations, not statistical



Perturbations: Temporal SECEDED (comb.↔pipe.) | Structural HSIAO (flat↔hierarchical) | constraint (10ns↔5ns) | Guarantee (SECEDED/HSIAO↔BCH, $t = 2$)

Fig. 1. Qualification precedes measurement. Code semantics admit a frozen hardware identity; constraints, tools, collateral, route seed, and activity then define a physical identity whose feasible evidence may be compared.

TABLE I
QUALIFIED IDENTITY AND EXPERIMENT MATRIX. W1/W2 DENOTE ERROR WEIGHT.

Hardware identity	Bits	Guarantee	Lat.	II	Qualification	Campaign	Feasible
SECEDED combinational	72	W1 corr.; W2 detect	1	1	exact temporal pair	10/5 ns, 5 seeds	5/5; 5/5
SECEDED pipelined	72	W1 corr.; W2 detect	3	1	exact temporal pair	10/5 ns, 5 seeds	5/5; 5/5
Hsiao flat	72	W1 corr.; W2 detect	1	1	arbitrary-word SAT	10 ns, 5 seeds	5/5
Hsiao hierarchical	72	W1 corr.; W2 detect	1	1	SAT + temporal induction	10 ns, 5 seeds	5/5
BCH (78, 64, $t = 2$)	78	W1/W2 correct	1	1	qualified guarantee	10 ns, 5 seeds	0/5

replicas or samples from a known population. No seed is discarded or replaced.

For a target T and signed worst setup slack s_{setup} , we report

$$f_{\text{slack}} = \frac{1000}{T - s_{\text{setup}}} \text{ MHz.} \quad (1)$$

This is a *post-route signed-slack-derived frequency estimate*, not measured F_{max} , a frequency sweep, or a re-optimized maximum-frequency implementation. Target feasibility separately requires nonnegative setup slack, zero setup violations, hold closure, route completion, and zero final DRC.

Power uses OpenSTA on final ODB/SDC/SPEF with a deterministic primary-input VCD propagated into each implementation. Each trace contains 100,000 useful no-error operations at $\text{II}=1$. Energy/op is reconstructed as total power times trace duration divided by useful operations; therefore power and energy are not interchangeable across the 10 and 5 ns traces. Missing or timing-ineligible measurements remain unavailable.

C. Evidence joins, pairing, and missingness

Every admitted numerical row is a join rather than a nearest-name lookup. The join key contains the frozen RTL/source inventory, top-level interface and transaction contract, target period, library/corner and flow configuration, route seed, final implementation database, parasitic file, and activity trace. Correctness evidence joins at hardware identity; route and power evidence additionally join at physical identity. A proof for one RTL cannot qualify a nominally similar netlist, and power from one routed database cannot be displayed beside timing or area from another. Source, configuration, and artifact hashes are retained at full precision even though the paper rounds display values.

The unit of analysis follows the claim. One physical point is one hardware identity at one seed and condition. For a paired comparison and metric x , the seed-level effect is

$$\Delta_s(x) = 100 \frac{x_{\text{changed},s} - x_{\text{baseline},s}}{x_{\text{baseline},s}}. \quad (2)$$

Pairing is applied before aggregation. We preserve all five Δ_s values, then report their mean, sample standard deviation, and min-max envelope descriptively. The seed is not a fabricated-chip replicate and the envelope is not a confidence interval. It tests the narrower question of whether a sign or ordering depends on one deterministic heuristic trajectory. The same seed simultaneously perturbs several metrics, but it is not treated as a one-dimensional “goodness” variable.

The campaign matrix, seeds, feasibility gates, targets, and metric rules were fixed before the admitted measurements. Completed target-infeasible routes retain area, routing, DRC, hold, signed setup slack, and violation count. They do not acquire target-clock energy merely through Eq. (1). No seed is removed for an unfavorable result and no identity receives a seed-specific density, floorplan, or synthesis recipe. An infrastructure retry would have to reuse the seed and preserve the failed attempt; the admitted matrices required no outcome replacement. These rules distinguish an unavailable measurement from a zero and prevent downstream analysis from repairing feasibility after observing PPA.

The three comparisons answer different questions. Temporal SECEDED changes register placement and latency while preserving the transaction relation after alignment. Structural Hsiao holds latency and II fixed while changing the decoder representation. Constraint replication holds both SECEDED RTL identities fixed while changing the target and period-dependent I/O delays. BCH changes guarantee and hardware identity together, so it is used only as a feasibility boundary. Only the first two comparisons support exact-equivalence implementation effects; none is a code-family ranking.

IV. RESULTS

A. Temporal restructuring produces a large mixed trade-off

At 10 ns, pipelining changes standard-cell area by 37.2% (36.7–37.8%), detailed wire by 11.2%, f_{slack} by 46.6% (42.1–55.3%), and energy/op by -23.4% (-24.0% to -22.5%). The directions hold in 5/5 matched seeds. The pipeline exchanges

higher area and three-cycle request latency for more timing margin and lower steady-stream energy; II remains one. With area, latency, timing, and achievable energy admitted, every seed and the aggregate means are non-dominated.

The absolute means establish scale. Combinational versus pipelined SECCED uses 18650 versus 25588 μm^2 and 44917 versus 49964 μm of detailed wire. The signed-slack estimate changes from 238.9 to 350.0 MHz, while equal-work energy changes from 145.77 to 111.63 pJ/op. Both close setup, hold, route, and final DRC in all five 10 ns runs. These are means of five closed physical identities, not one route replicated five times or values transplanted from a historical result.

Variability is metric-dependent. The area effect has a sample standard deviation of 0.4 percentage points and a narrow 36.7–37.8% envelope; detailed wire spans 10.6–12.1% with 0.6 points of descriptive spread. Energy has 0.6 points of spread. Timing is directionally stable but has 5.1 points of spread and spans 42.1–55.3%. Thus the seed evidence strengthens the sign of the temporal trade-off inside this closed set while showing that its timing magnitude is the most trajectory-sensitive of the four headline effects.

B. Structural identity matters without a large effect

Figure 2 deliberately preserves the scale contrast. Hierarchical relative to flat Hsiao changes area by -0.564%, cells by +0.276%, detailed wire by +3.867%, vias by +3.918%, f_{slack} by +2.656%, and energy/op by +1.328%. Area is lower in 5/5 seeds; cells, wire, vias, and energy are higher in 5/5; timing is higher in 4/5. This is neither “hierarchical Hsiao is worse” nor “flat Hsiao is superior”: area and timing move favorably on average while routing and energy move unfavorably, and one timing ordering reverses.

The envelopes make the modest effect auditable. Area ranges from -0.877 to -0.249%, wire from +1.783 to +8.304%, and energy from +0.144 to +2.497%. Timing spans -5.887 to +13.106%, including the one reversal. The mapped difference is real but small: ten additional cells, unchanged register count, seven fewer XOR/XNOR cells, and different type histograms and name-independent graph signatures. Because the shared synthesis report did not expose combinational-depth paths, depth remains not assessable rather than being inferred from timing or cell counts.

The flat identity’s measured absolute means are 19191 μm^2 , 212.9 MHz, 49810 μm detailed wire, and 149.57 pJ/op, with 10 ns feasibility in 5/5 runs. Exact semantic and temporal identity therefore does not eliminate physical displacement, but neither does it prescribe its magnitude. Implementation identity tells us *what must be measured separately*, not that every implementation transformation must produce a large difference.

C. Power decomposition explains the temporal energy difference

At 10 ns, mean internal power is 6.7533 versus 7.2104 mW, switching is 7.8221 versus 3.9514 mW, leakage is 7.3003 versus 9.8134 nW, and total is 14.5754 versus 11.1618 mW.

Pipelining changes these components by +6.77%, -49.48%, +34.43%, and -23.4% total/energy, respectively. The switching decrease is the largest absolute mean top-level contribution and is sufficient to overcome increased internal and leakage power.

Cell-group totals reinforce the accounting: combinational-group power changes by -55.50%, whereas sequential and clock groups change by +70.86% and +74.34%. This decomposition identifies numerical contributors, not a preserved causal wave-form mechanism. Reduced glitch propagation is plausible, but per-net activity was not retained sufficiently to establish it.

This distinction matters because several stories fit the aggregates. Added registers and clock-tree cells explain why sequential and clock groups rise, but they do not by themselves prove why downstream switching falls. Placement, buffering, logic restructuring, and the applied no-error stream are all part of the physical identity. The evidence supports a component accounting—the 3.8707 mW mean switching reduction exceeds the 0.4571 mW internal increase and nanowatt-scale leakage increase—and a bounded hypothesis of glitch suppression. It does not support a per-net causal attribution or a general rule that pipelining reduces codec energy.

D. Constraint tightening changes magnitude, not ordering

All ten fresh 5 ns implementations are route-complete, DRC-clean, hold-clean, setup-feasible, and power-eligible; both identities meet the target in 5/5 seeds. Pipeline relative to combinational changes area by +36.7%, cells +27.9%, wire +12.8%, vias +18.6%, f_{slack} +68.6%, and energy/op -19.3%. Area, timing, power, and energy directions match 10 ns in every seed. The energy advantage narrows from about 23.4% to 19.3%, while the timing-margin advantage grows.

The 5 ns area effect spans +36.0 to +37.5%, the timing effect +63.6 to +75.9%, and energy -20.2 to -18.0%. Consequently, the preserved signs are not artifacts of averaging across a reversal. The test is deliberately harder than merely recomputing Eq. (1) from the 10 ns routes: each point is a fresh implementation under the new target, so target-driven synthesis, buffering, placement, and routing may change. Conversely, it is not a voltage/frequency-scaling experiment because voltage, library, corner, RTL, useful operations, and flow remain fixed.

Across physical conditions, combinational and pipelined energy/op change by -5.48% and -0.45%, while total power changes by +89.0% and +99.1%. The trace clock period is halved, so the near doubling of power does not imply doubled energy/op. This is constraint-condition robustness within the same technology/library/corner, not technology portability.

E. Stronger-correction feasibility boundary

The evaluated BCH syndrome/Chien RTL has mean area 57240 μm^2 , f_{slack} 66.6 MHz, detailed wire 185035 μm , signed setup slack -5.01 ns, and 66 setup violations per seed. It meets 10 ns in 0/5 runs. Its W1/W2-correction guarantee is stronger than SECCED/Hsiao, so it cannot be replaced by a weaker code merely because the physical point misses timing. Conversely, this one result does not show that BCH is inherently slow or infeasible; it shows only that this qualified realization misses the common target.

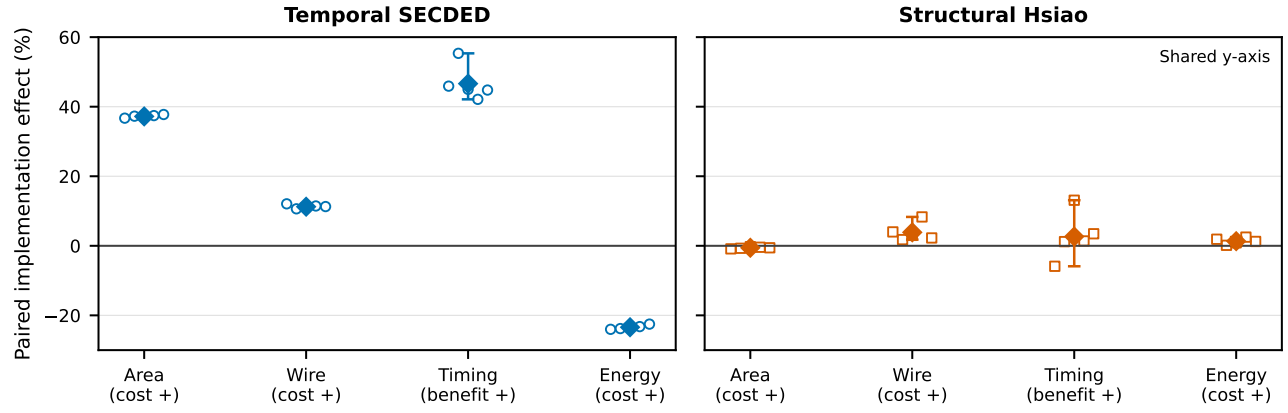


Fig. 2. Individual seeds (open markers), means (filled diamonds), and min-max ranges for the two exact-equivalence studies. Both panels intentionally share one y-axis: temporal SECEDED restructuring causes a large displacement, while structural Hsiao refactoring causes a modest mixed displacement. Positive area/wire/energy is cost; positive timing is benefit.

TABLE II

HEADLINE POST-ROUTE RESULTS. PAIRED EFFECTS ARE CHANGED RELATIVE TO BASELINE; VALUES ARE MEANS OVER FIVE MATCHED DETERMINISTIC SEEDS.

Study	Area	Timing metric	Energy/op	Ordering	Interpretation
Temporal SECEDED @ 10 ns	+37.2%	+46.6%	-23.4%	area/timing/energy 5/5	Large non-dominated area-latency-timing-energy displacement.
Structural Hsiao @ 10 ns	-0.564%	+2.656%	+1.328%	area/energy 5/5; timing 4/5	Modest mixed displacement; neither structure is generally superior.
Temporal SECEDED @ 5 ns	+36.7%	+68.6%	-19.3%	all three 5/5	Ordering survives; magnitudes remain constraint-conditioned.
BCH @ 10 ns	57240 μm^2	66.6 MHz	—	setup feasible 0/5	Stronger guarantee; this syndrome/Chien RTL misses the common target.

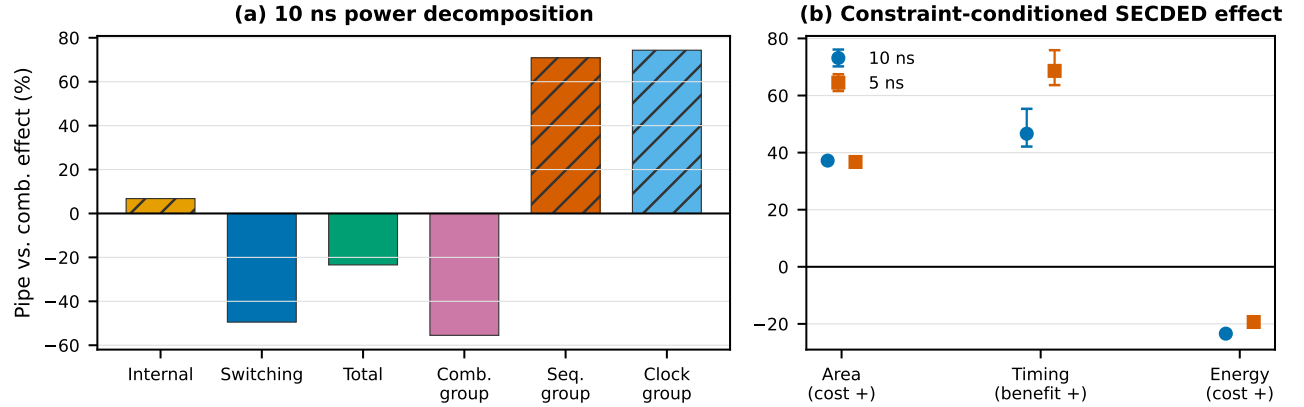


Fig. 3. (a) Mean paired SECEDED power effects at 10 ns: switching and the combinational-cell group dominate the reduction despite sequential/clock increases. (b) Temporal-pair effects at 10 and 5 ns. Individual-seed min-max bars show ordering preservation with condition-dependent magnitude.

V. DISCUSSION AND LIMITATIONS

The two exact-equivalence experiments give the main methodological result. Temporal restructuring displaces registers, latency, switching, and routing enough to create a large non-dominated trade-off. Non-temporal Hsiao restructuring changes mapped logic and physical outcome much less. The contrast is scientifically useful: semantic identity requires separate physical measurement, but does not predict either effect direction or magnitude. Matched seeds test whether an ordering is a one-trajectory artifact; five deterministic seeds do

not support p-values, confidence intervals, tail probabilities, or population inference.

A. Reading large and small implementation effects

The temporal and structural comparisons should not be read as competing demonstrations of the same perturbation strength. The SECEDED pair changes the placement of state across the datapath and therefore changes request latency, register and clock burden, logic available per stage, and switching propagation together. Its large effects are coherent with a broad architecture transformation, but the data do not decompose

that transformation into independent causes. The Hsiao pair is deliberately tighter: encoder, syndrome matrix, transaction boundaries, state count, latency, and II are held fixed, and only the decode organization changes. A sub-percent area shift and low-single-digit routing, timing, and energy shifts are a credible outcome of that narrower intervention.

Small does not mean physically nonexistent. The Hsiao area sign is consistent, as are the opposing wire, via, and energy signs, while timing reverses once. The result rejects two convenient simplifications: formal equivalence does not imply identical physical outcome, and structural distinctness does not imply a practically large displacement. Reporting the panel on an expanded axis would conceal the experimental comparison that matters. The common scale instead shows why implementation identity is a measurement rule rather than an effect-size prediction.

The two studies also have different decision consequences. For temporal SECEDED, latency and area prevent the energy/timing-favorable pipeline from dominating; a one-cycle service requirement may settle the choice before PPA optimization. For structural Hsiao, categorical requirements do not distinguish the pair, so the modest mixed effects can matter only near an engineering threshold or after aggregation across many instances. A designer should not infer a universal preference from the mean: flat is favorable for cell count, wire, vias, and energy here, while hierarchical is favorable for area and mean timing. The single timing reversal further argues for retaining the seed envelope when timing margin is tight.

B. Portability and decision boundaries

Implementation condition also belongs in physical identity. A $2\times$ target tightening preserves the temporal ordering here but changes magnitudes. The method generalizes: freeze semantic and hardware identities, qualify behavior, join route/power evidence, preserve missingness, and compare matched perturbations. The numerical area, timing, energy, Hsiao direction, and BCH feasibility do not generalize across RTLs, tools, libraries, nodes, corners, or floorplans. This separates *method portability* from *numerical portability*.

Selection should therefore be staged. First filter by categorical guarantee, correctness status, request latency, and target feasibility; only then compare admitted hardware identities on physical objectives. A one-cycle requirement favors combinational SECEDED over its exact-equivalent pipeline; a strict W2-correction requirement admits BCH semantics but exposes an implementation gap. No scalar PPA score resolves either categorical requirement.

Quantitative scope is one open 130-nm library, one principal TT corner, specific RTL identities, one floorplan policy, two timing targets, and five heuristic seeds. Power is post-route activity-based estimation, not silicon measurement. We exclude SRAM macro integration, PVT and IR-drop sweeps, clock-domain integration, scrubbing/interleaving, physical fault adjacency, and FIT/SER claims. Equal-work no-error activity does not predict correction-heavy traffic, and common policy

reduces configuration confounding without proving architecture-specific optimal floorplans.

The strongest alternative explanations are correspondingly bounded. First, a different synthesis or physical flow may absorb the Hsiao representation difference or magnify it; the present result establishes neither invariance nor insignificance. Second, the temporal effect combines register insertion, latency, retiming opportunities, clock distribution, and activity redistribution, so exact equivalence identifies the changed architecture as a whole rather than a single causal gate class. Third, a macro-integrated memory system may value decoder latency, correction traffic, and physical adjacency differently from the steady streaming codec boundary used here. These are limits on numerical transfer, not defects in the joined evidence for the frozen identities.

Reproducibility is therefore claim-oriented. A replication should retain the same semantic and transaction qualification, predeclare its physical conditions and perturbations, preserve all completed and ineligible points, and regenerate paired claims from raw manifests. A different sign under another node or flow would be informative: it would belong to a new physical identity. A failed proof or broken provenance join, by contrast, would make the comparison unavailable rather than contradictory.

The evidence hierarchy also limits the novelty claim. Exact proofs establish behavior for the named RTL pairs; matched routes establish within-flow displacement and sign robustness; component totals establish an accounting; the second target establishes one-condition replication. None alone establishes a causal microarchitectural mechanism, a globally optimized layout, or a family-wide ranking. The contribution is the joined experimental argument and the contrast it exposes, not a claim that any constituent proof, seed control, power estimator, or open flow is itself new.

VI. CONCLUSION

ECC semantics, hardware architecture, and physical implementation are distinct experimental identities. Exact qualification plus matched physical evidence reveals a large temporal SECEDED trade-off, a modest mixed Hsiao structural displacement, switching-dominated temporal energy reduction, and constraint-conditioned magnitude with stable ordering. Stronger correction remains a categorical feasibility boundary. The appropriate comparison is therefore between qualified, feasible, explicitly named implementations—not family labels.

VII. ACKNOWLEDGEMENT

Generative AI Disclosure: OpenAI ChatGPT (model sol 6) was used to assist in drafting and language refinement of limited portions of the Introduction and Discussion sections. All AI-assisted material was subsequently reviewed, revised, and verified by the authors. The tool was not used to generate experimental data, RTL implementations, formal-verification results, physical-design measurements, or scientific conclusions.

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